Week 4. By Deng Yufan.

Prob 1. Give a model for $\phi_{Lt} = \forall x [R_1(x,x)] \wedge \forall x,y [R_1(x,y) \leftrightarrow R_1(y,x)]$
$\wedge \forall x,y,z [(R_1(x,y) \wedge R_1(y,z)) \rightarrow R_1(x,z)]$
$\wedge \forall x,y [R_1(x,y) \rightarrow \neg R_2(x,y)]$
$\wedge \forall x,y [\neg R_1(x,y) \rightarrow (R_2(x,y) \oplus R_2(y,x))]$
$\wedge \forall x,y,z [(R_2(x,y) \wedge R_2(y,z)) \rightarrow R_2(x,z)]$.
$\wedge \forall x \exists y [R_2(x,y)]$.

~~We can let $U = \mathbb{Z}$, $R_1(x,y) = 2 \mid (x-y)$, $R_2(x,y) = [x \text{ xor } y = 1]$~~.

We can let $U = R$ (real number set), $R_1(x,y) = [[x] = [y]]$, $R_2(x,y) = [[x] < [y]]$.

where $[x]$ is maximal integer not larger than x.

Easy to verify all conditions.

Prob 2. Prove $HALT_\varepsilon = \{\langle M \rangle \mid M \text{ halts on empty input}\} \notin R$.

If $HALT_\varepsilon \in R$. then $\exists$ TM T that accept $HALT_\varepsilon$ and always halt.

Consider a TM M: For any input x,

Step 1. Get its own code $\langle M \rangle$.

Step 2. Simulate T with input $\langle M \rangle$.

Step 3. If T accept. Loop. Otherwisle halt.

So if T accept $\langle M \rangle$, $\langle M \rangle \notin HALT_\varepsilon$. if reject, $\langle M \rangle \in HALT_\varepsilon$. contradiction

Prob 3. Prove any infinite subset of $MIN_{TM} = \{\langle M \rangle \mid M \text{ is a minimal TM}\}$

is not in RE.

If $\exists S \subseteq MIN_{TM}$. S infinite. $S \in RE$. then $\exists$ TM T that accept S.

Consider NTM T' with input n:

Step1. Guess a string w longer than n.

Step 2. Simulate T with input w.

Step 3. If T halts, ~~clear the tape~~ write w on tape and clear other parts. then halt.
Other wise never halt.

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Then consider TM M: For any input x.

Step 1. Get its own code $\langle M \rangle$.

Step 2. Count length of $\langle M \rangle$, n.

Step 3. Simulate T' with input n. Since S infinite, T' always halt.

Step 4. Simulate the output of T', w with input x.

So $w \in S$, but $\langle M \rangle$ is shorter than w and equivalent to w.
contradiction.

Prob 4. a. Prove $\exists$ Computable $S_1' : \Sigma^* \times \Sigma^* \to \Sigma^*$, which turn TM $\langle T \rangle$ and input x into TM $\langle S \rangle$ that

$S(y) = T(x, y)$ if T halt with $\langle x, y \rangle$.

$S(y)$ loop otherwise.

Let $q(s)$ be a computable function, returns a TM $\langle Q \rangle$ that for any input x, Q halt with only $s\#x$ on tape.

Consider TM S with input $\langle T \rangle$ and x:

Step 1. Calc $\langle Q \rangle = q(x)$.

Step 2. Output combination of $\langle Q \rangle$, $\langle T \rangle$.

Obviously S satisfy the condition.

b. Prove Kleene's recursion thm by a. and Roger's fixed-point thm.

~~For all TM T we can have $S'' : \Sigma^* \to \Sigma^*$.~~

For all TM $T : \Sigma^* \times \Sigma^* \to \Sigma^*$, we can have computable $S'' : \Sigma^* \to \Sigma^*$ that

$S''(x) = S_1'(\langle T \rangle, x)$ by using a. to construct S_1' and S''.

Then we can have TM F that $S''(\langle F \rangle) = \langle F \rangle$ by using Roger's.

So $\langle F \rangle = S''(\langle F \rangle) = S_1'(\langle T \rangle, \langle F \rangle)$ thus
$T(\langle F \rangle, y) = F(y)$. So F is the solution for Kleene's.

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